

IUS
INSTITUT
UNIVERSITAIRE
DES SCIENCES

FALCUTÉ DES SCIENCES ET TECHNOLOGIES (FST)

TROISIÈME ANNÉE

Rapport du travail de Laboratoire NO 1

Cours : Réseaux

Étudiant : Rodney E.Sainto
Professeur : Ismaël SAINT AMOUR

Le 07 Novembre 2026

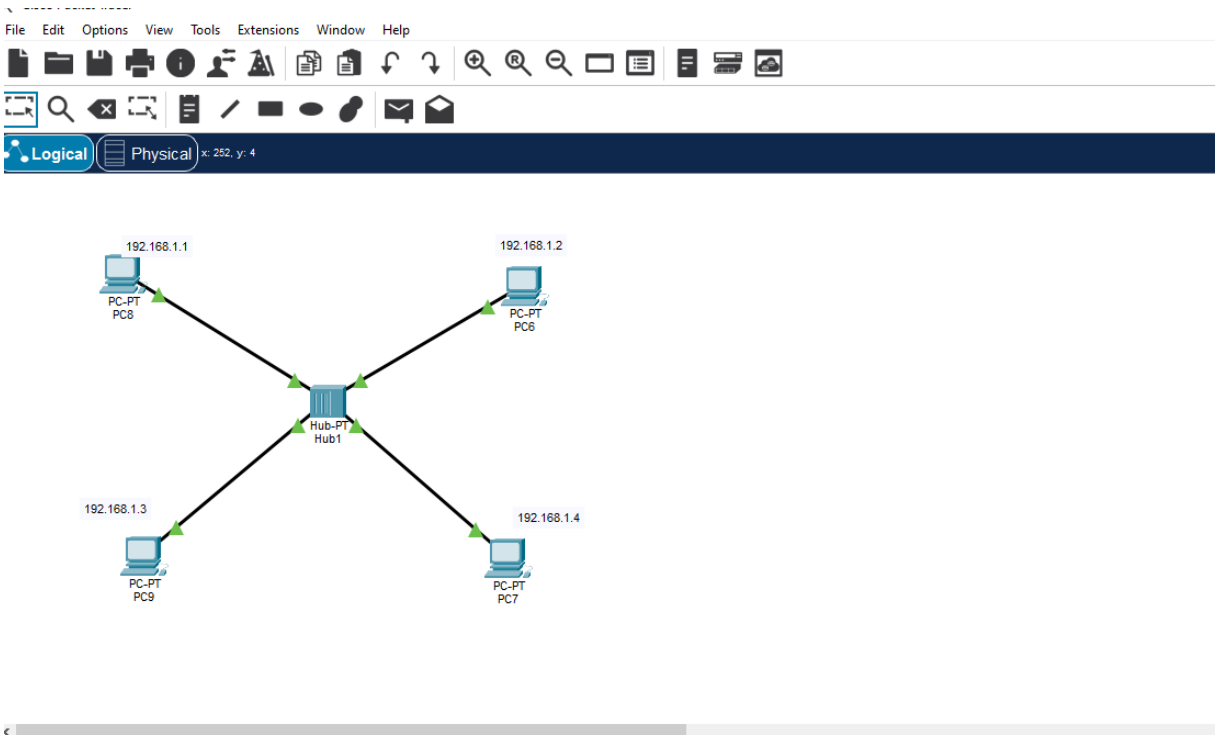
L’objectif de ce TD est de :

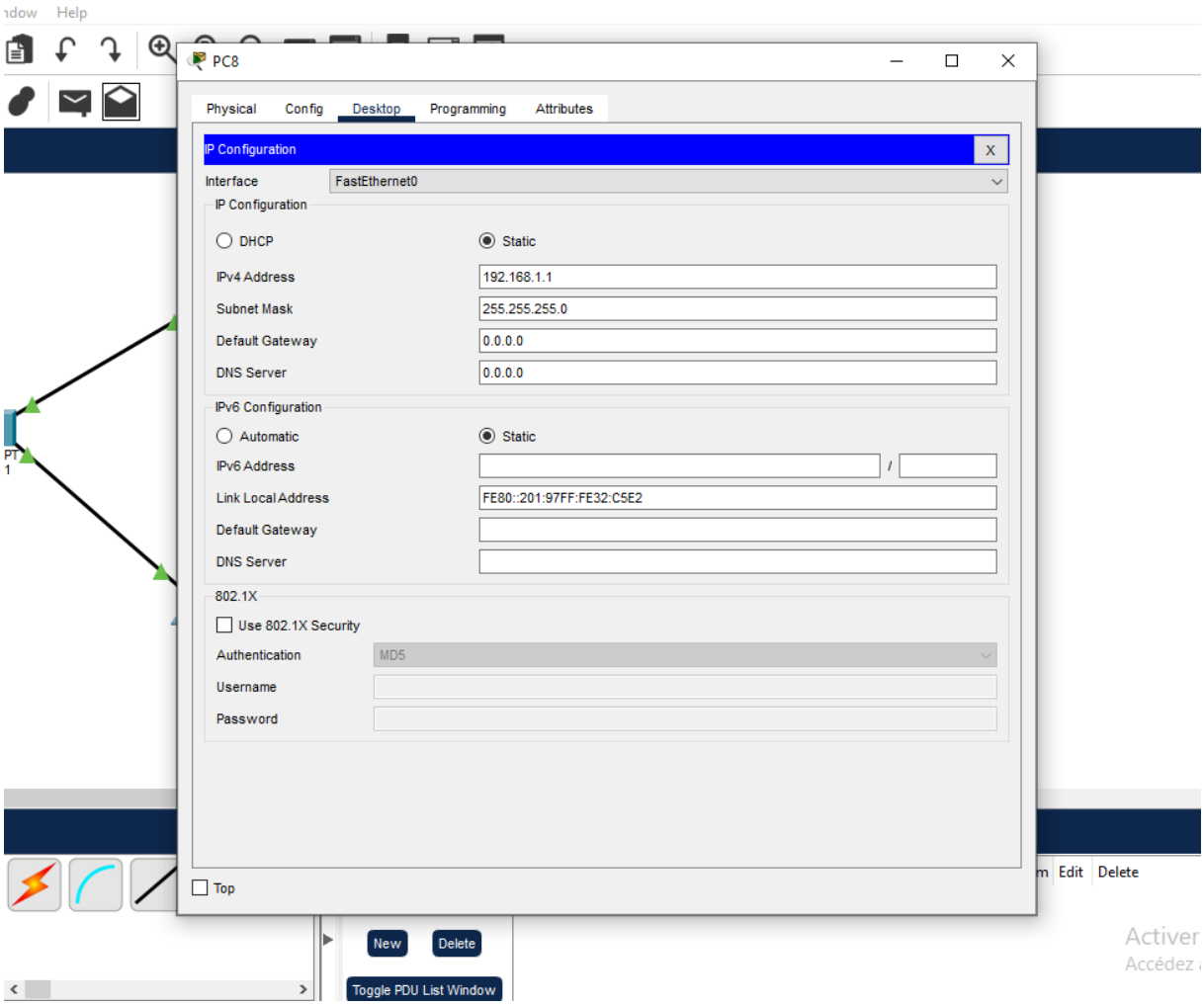
1. Savoir attribuer des adresses IP valides aux machines.
2. Comprendre l’adressage IPv4 et IPv6.
3. Configurer des adresses IP sur des hôtes et routeurs dans Cisco Packet Tracer.
4. Vérifier la connectivité avec les commandes ping et ping ipv6.

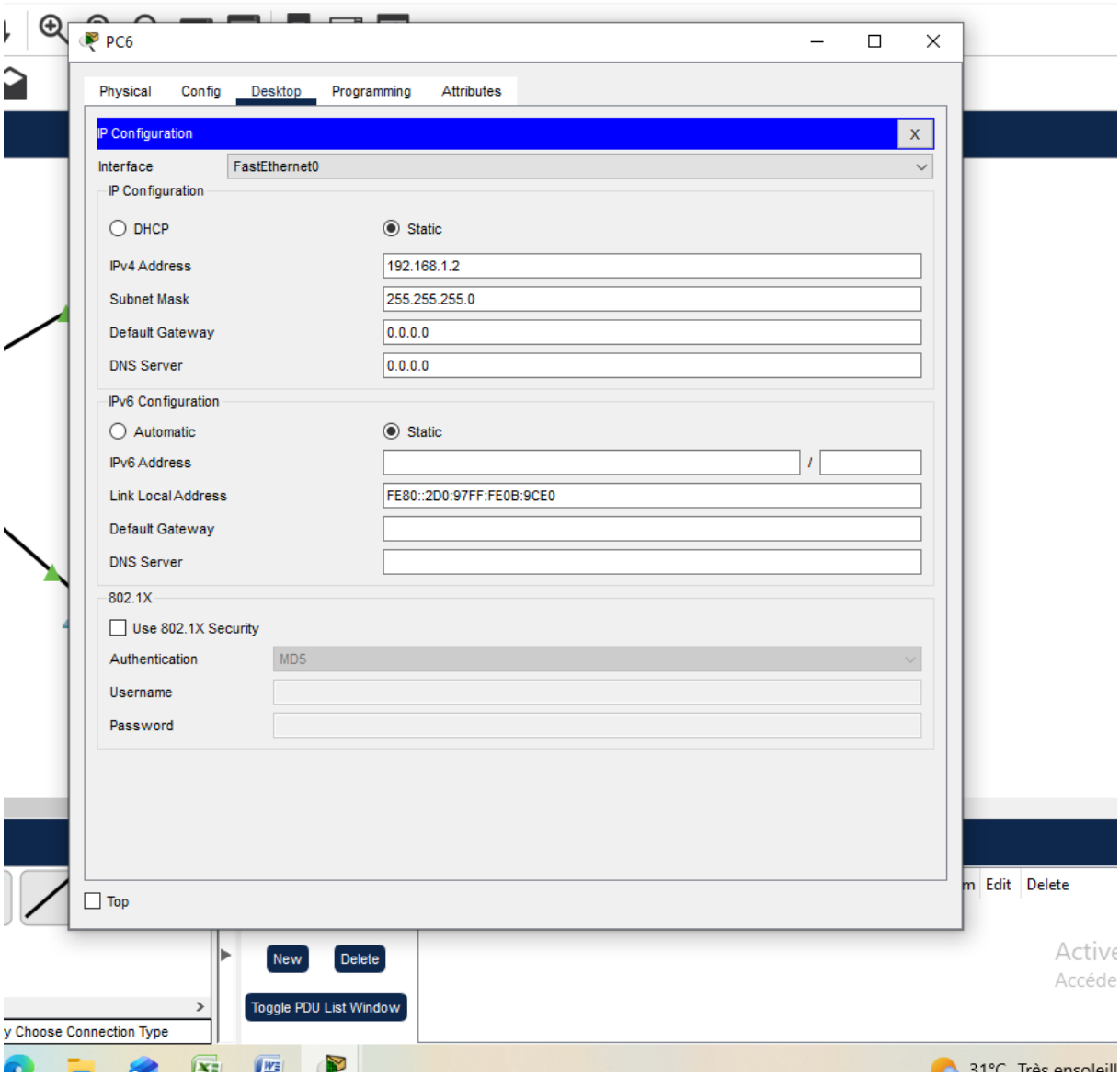
Travail Dirigé

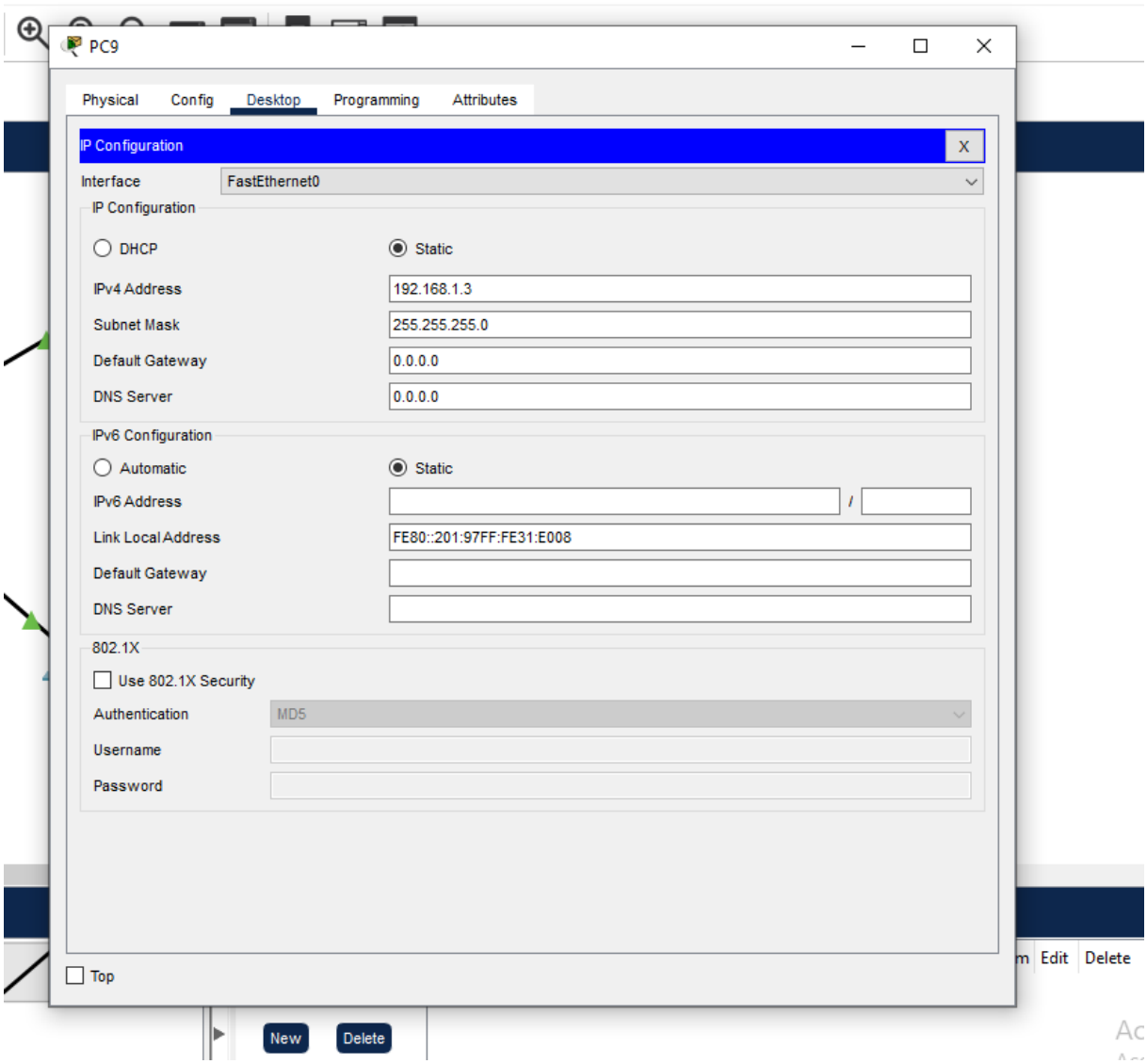
- Reproduire une série de tâches.
- Création une topologie
- Reproduire les topologies en configurant les adresses IPv4 et IPV6 indiquées dans le tableau ci-dessous, puis vérifiez la connectivité.

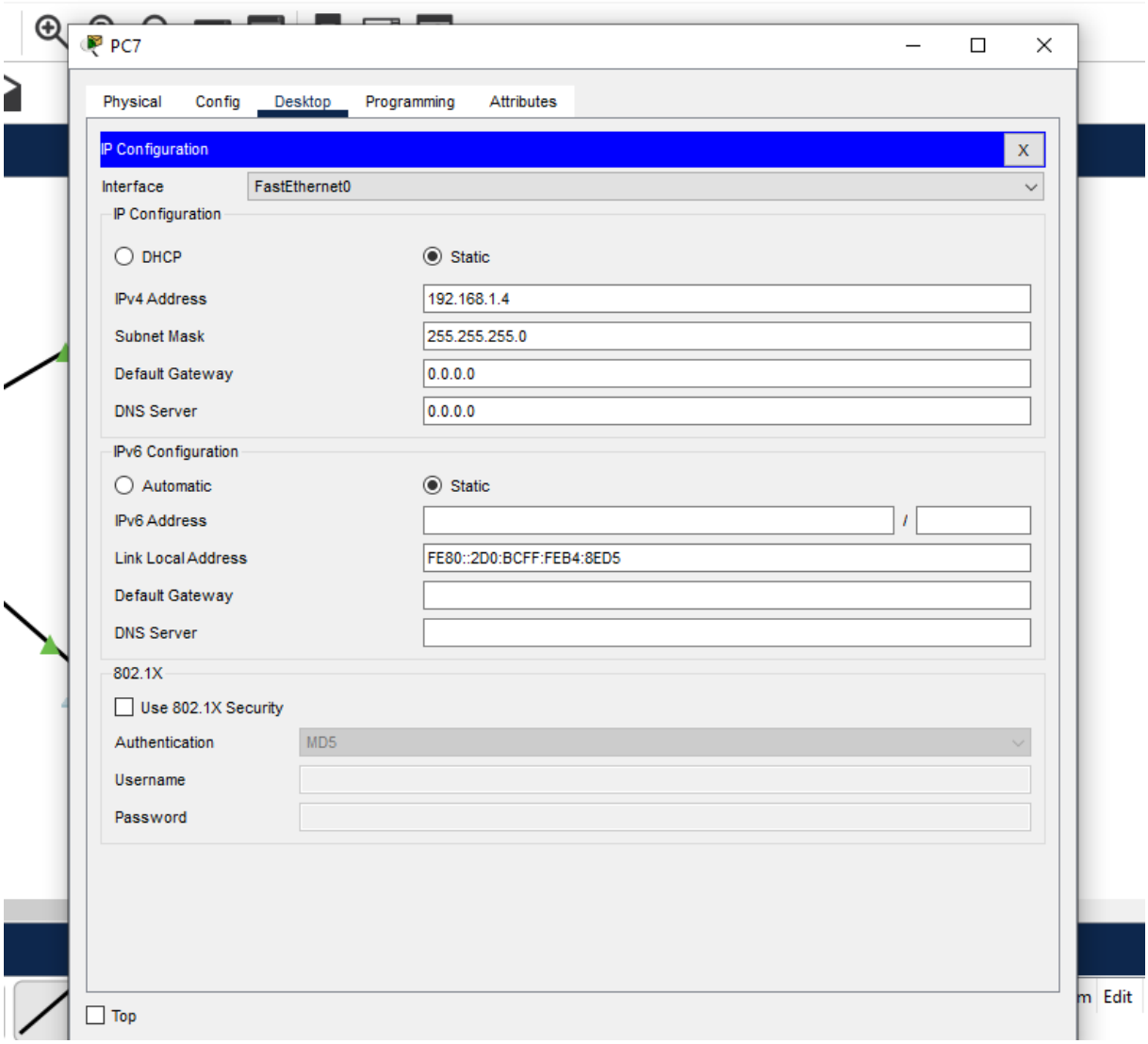
1. **Créez une topologie avec : Un Hub. 4 PC connectés au Switch avec IPV4.**



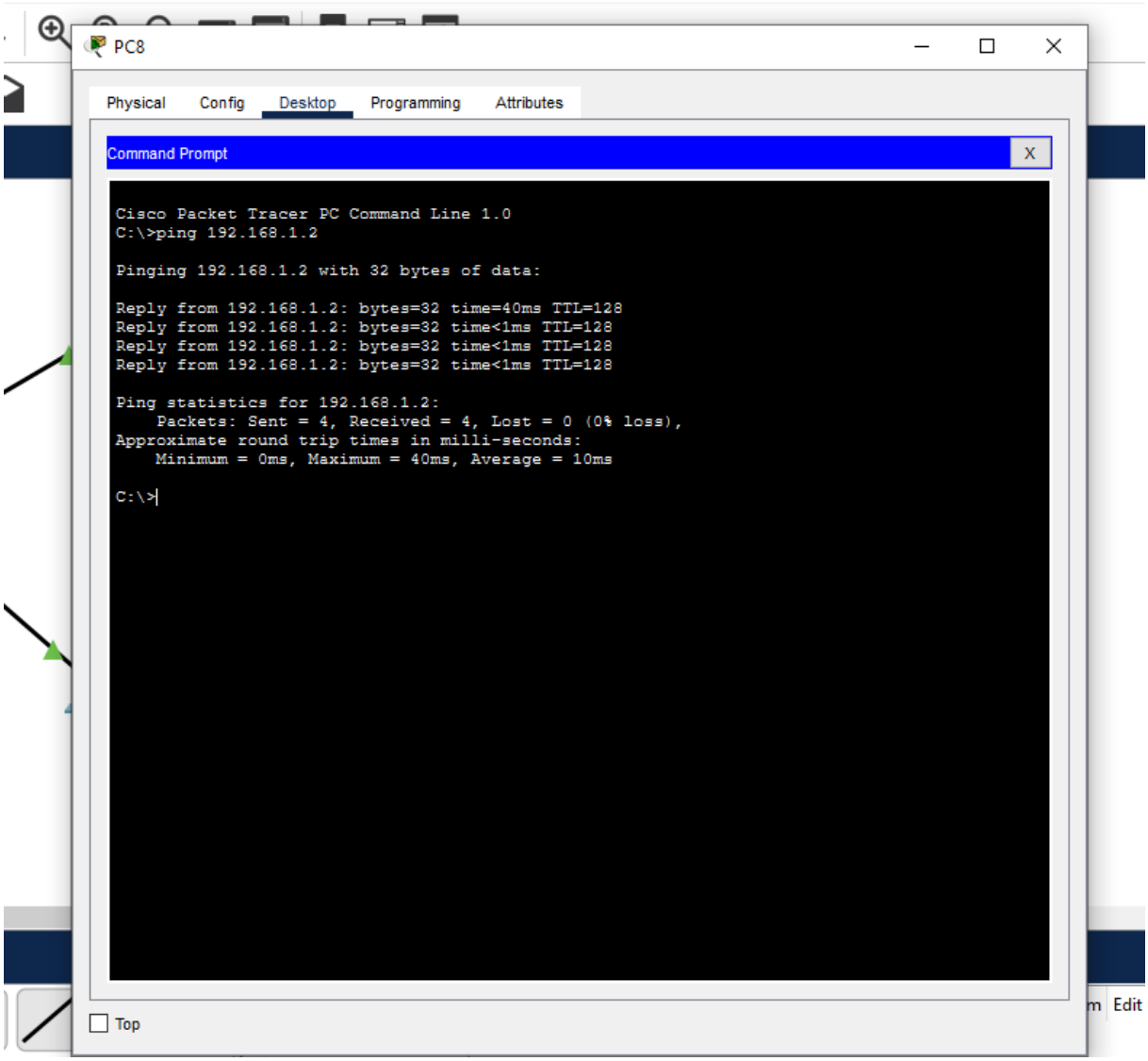

PC8
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.1
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::201:97FF:FE32:C5E2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Edit
Delete
New
Delete
Toggle PDU List Window


PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:97FF:FE0B:9CE0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Edit
Delete
New
Delete
Toggle PDU List Window
Choose Connection Type

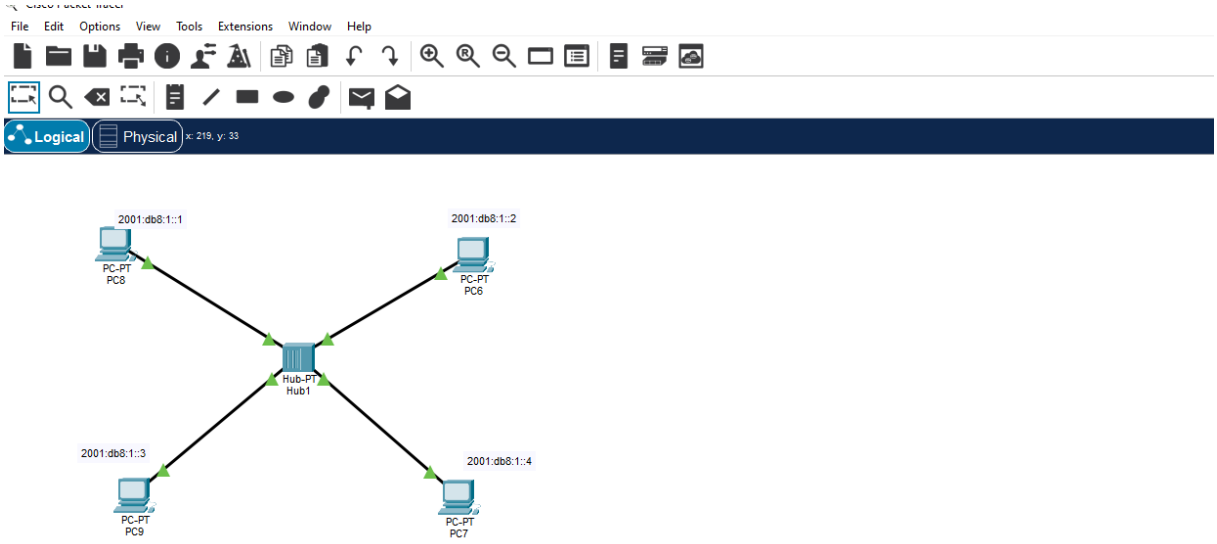

PC9
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.3
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:97FF:FE31:E008
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
New
Delete
Edit
Delete

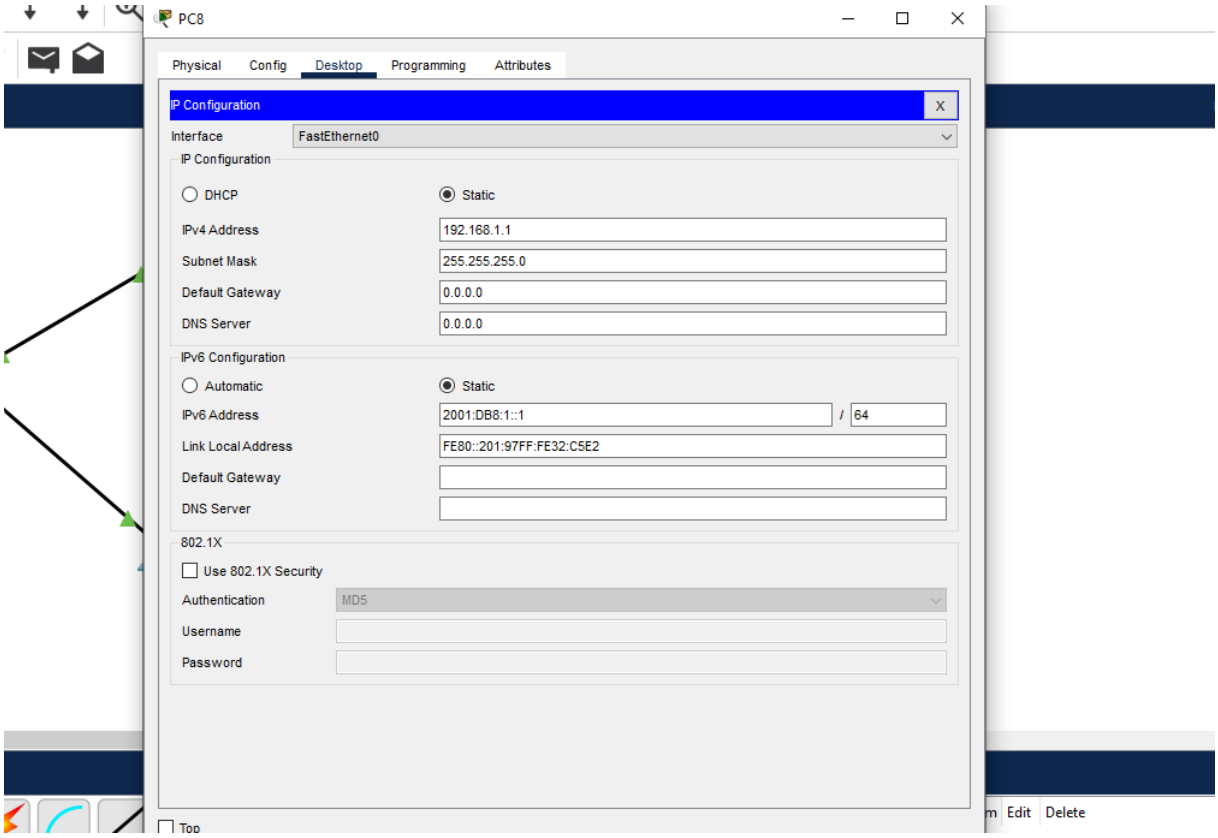

PC7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.4
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2D0:BCFF:FEB4:8ED5
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Edit

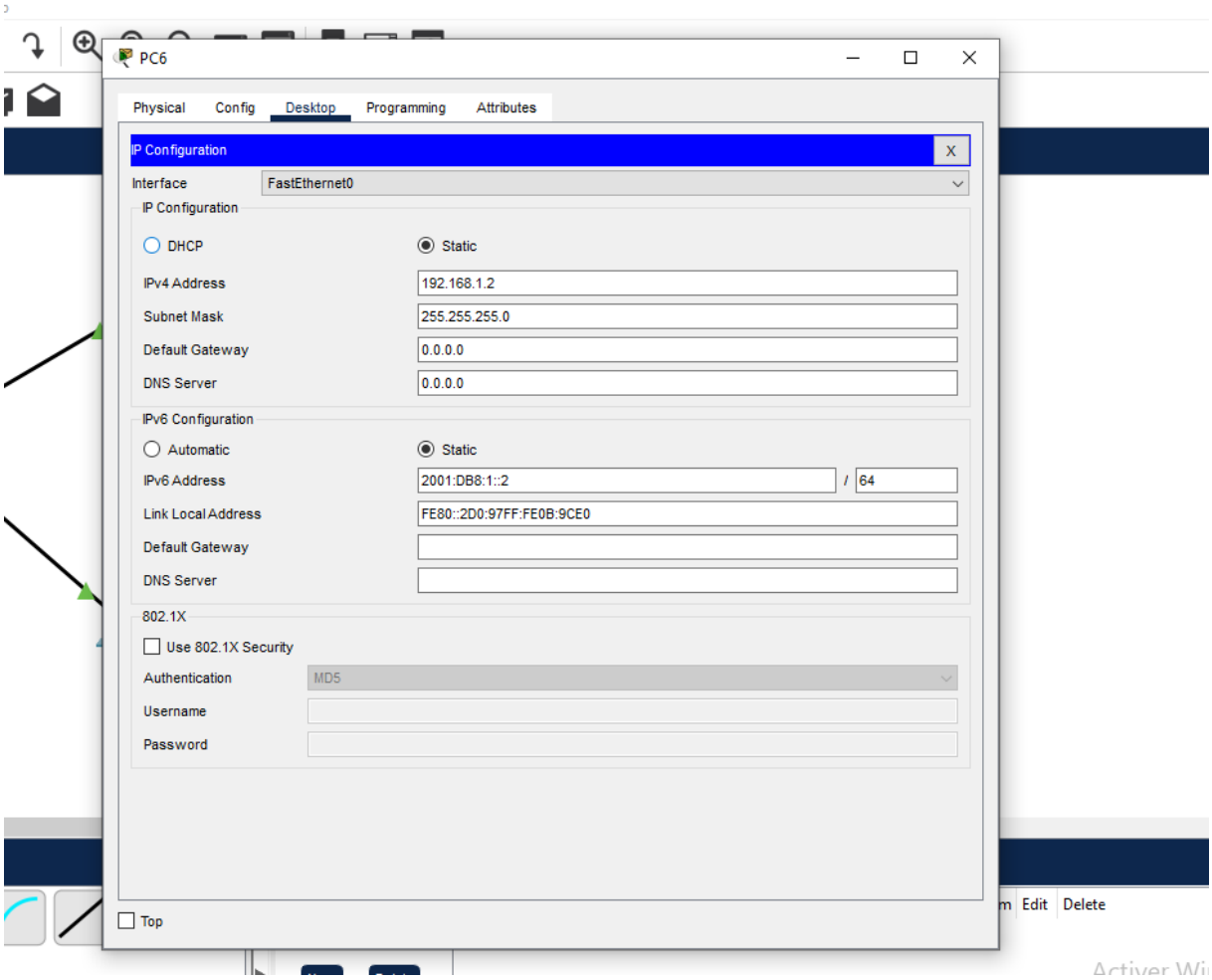
Testons la connectivité en utilisant la commande



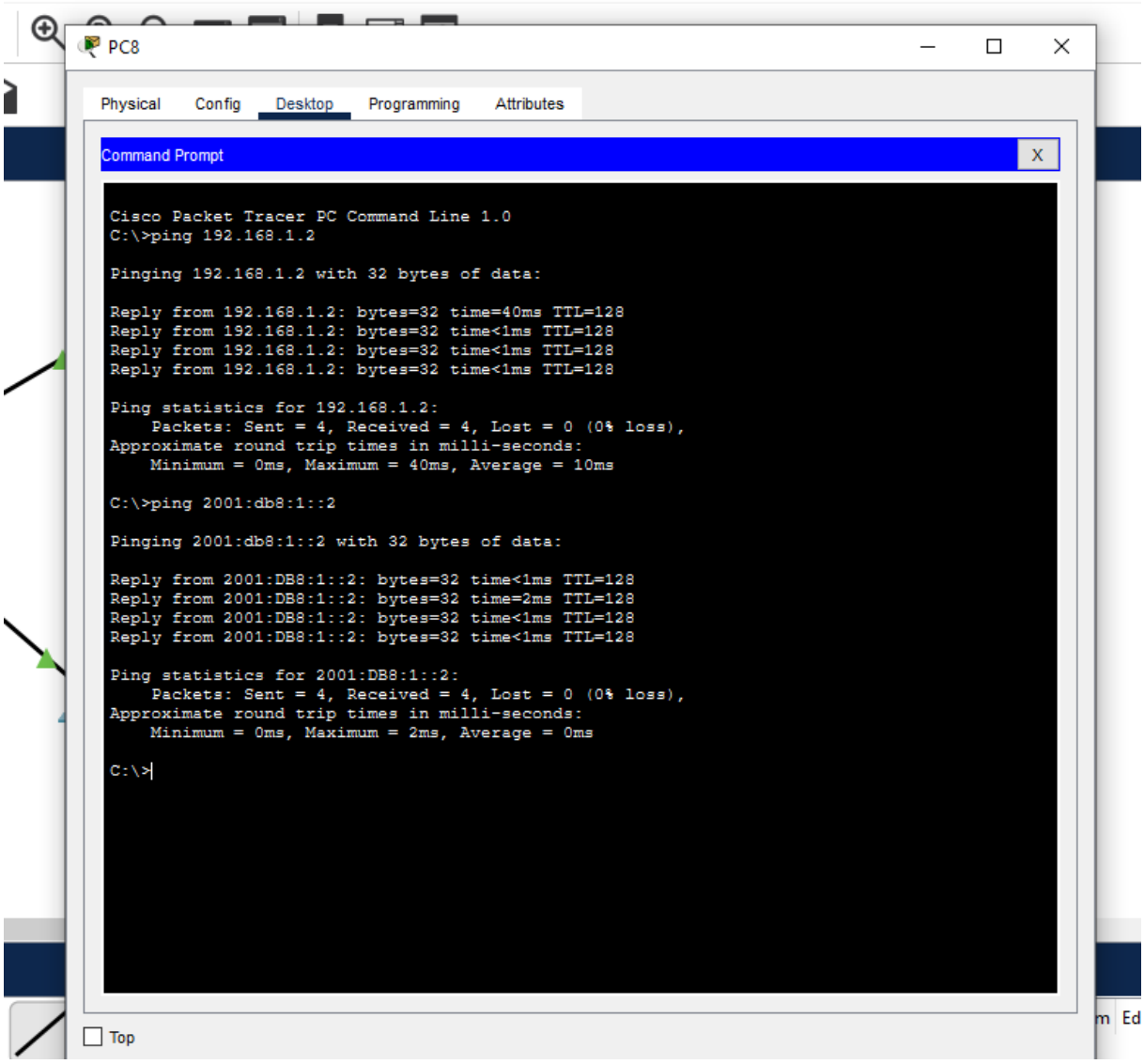
2. Dans la même topologie, configurez également une adresse IPv6 sur chaque PC.



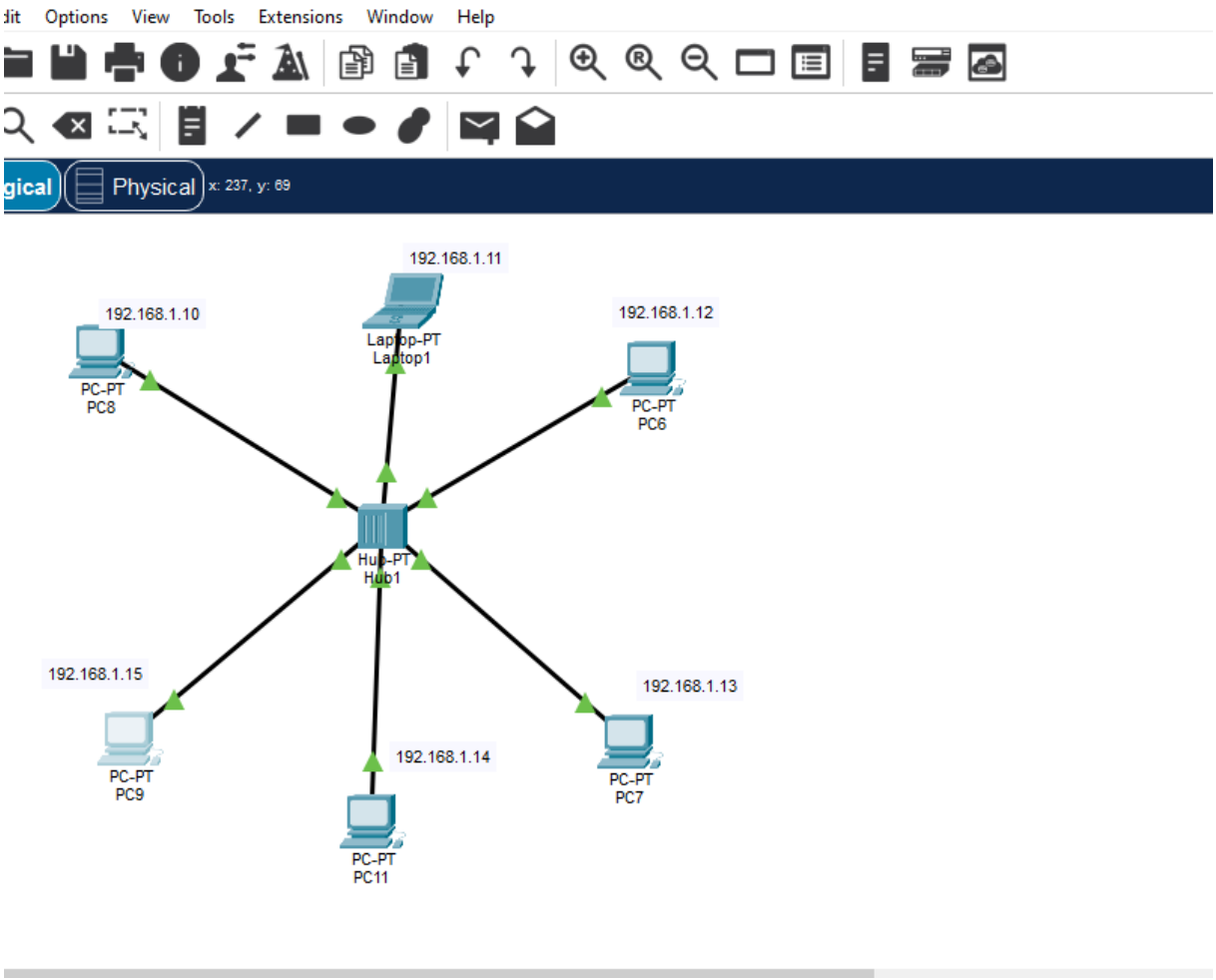


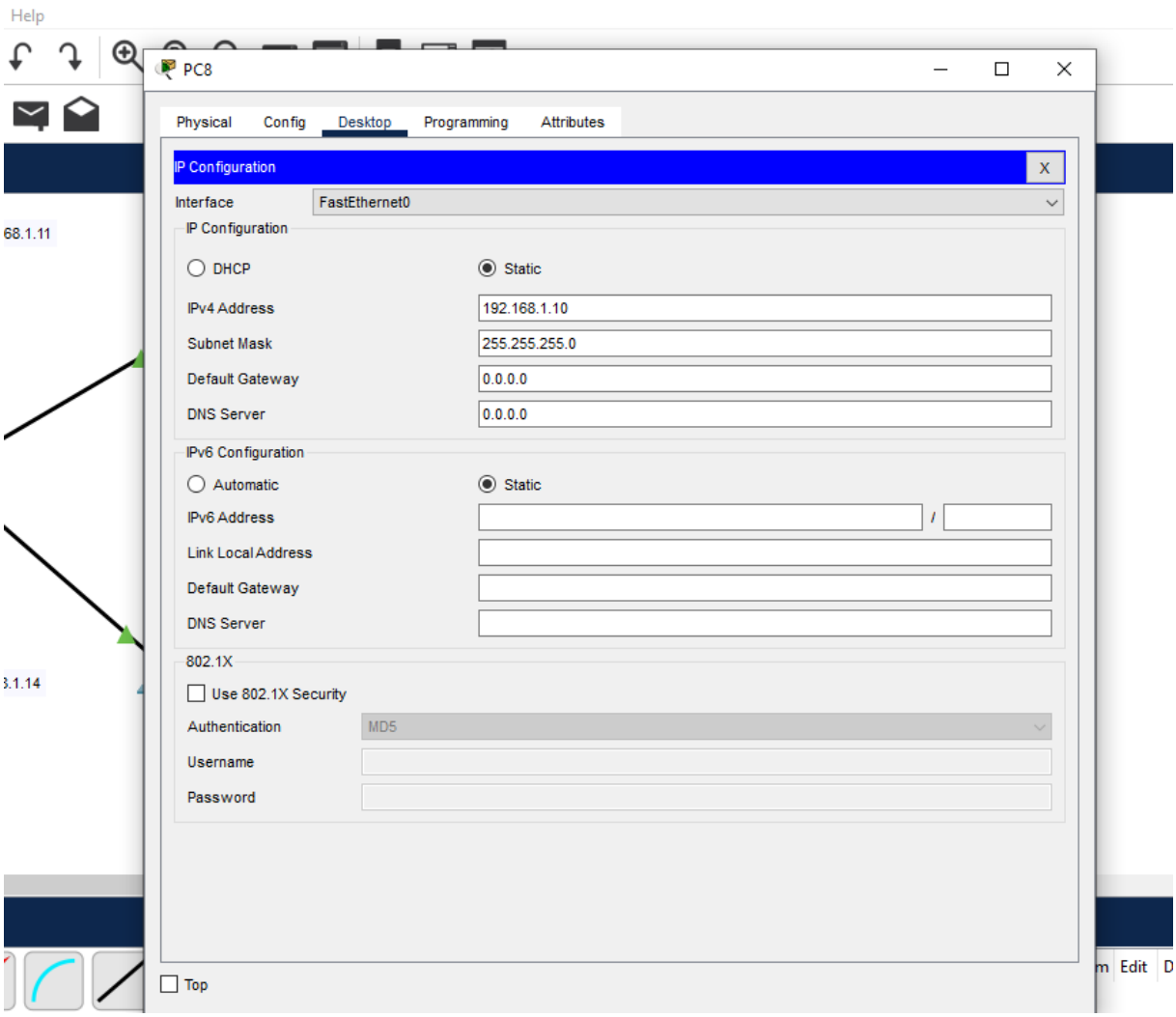

PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.2
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
2001:DB8:1::2
/
64
Link Local Address
FE80::2D0:97FF:FE0B:9CE0
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Edit
Delete

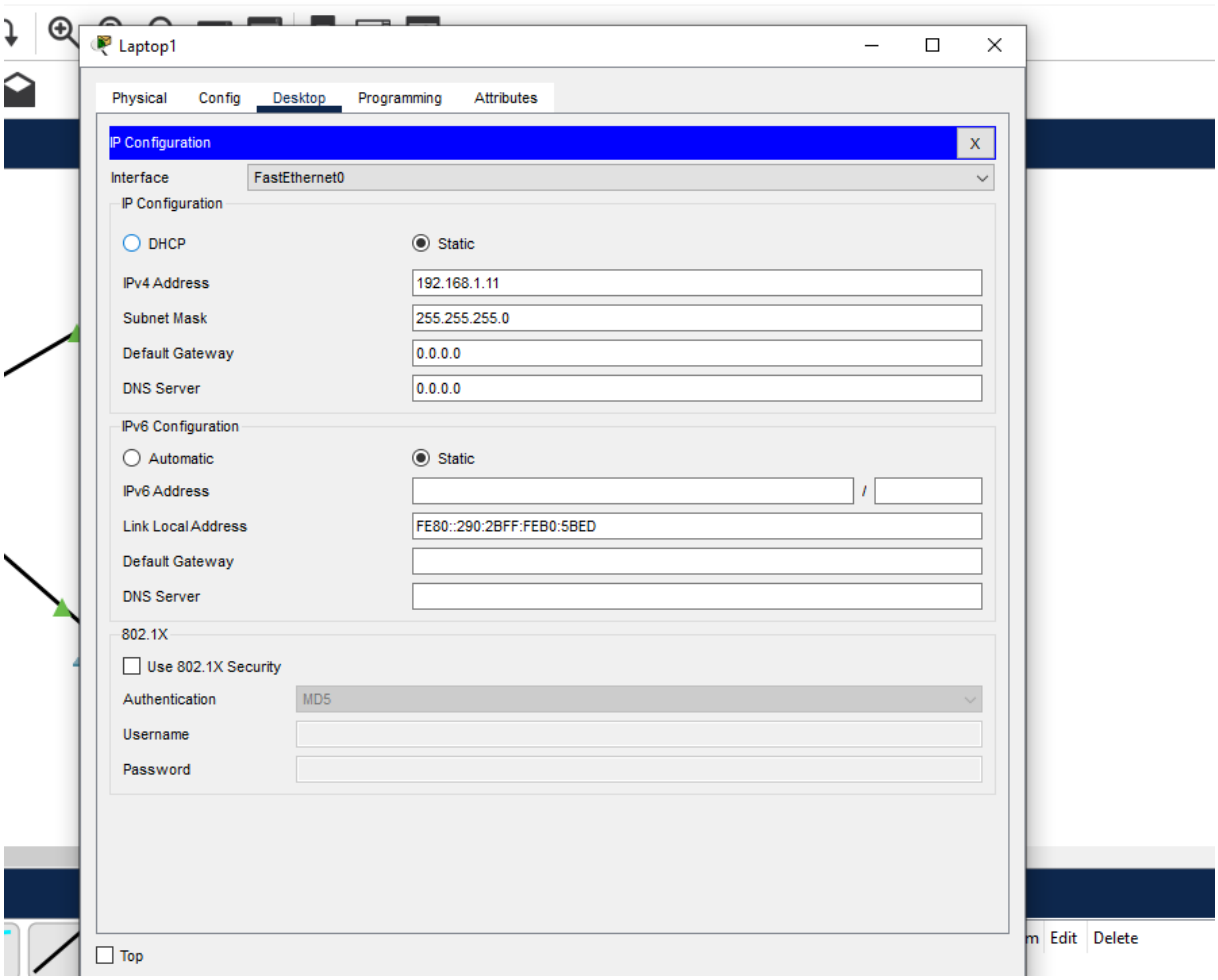
Testons la connectivité en utilisant la commande



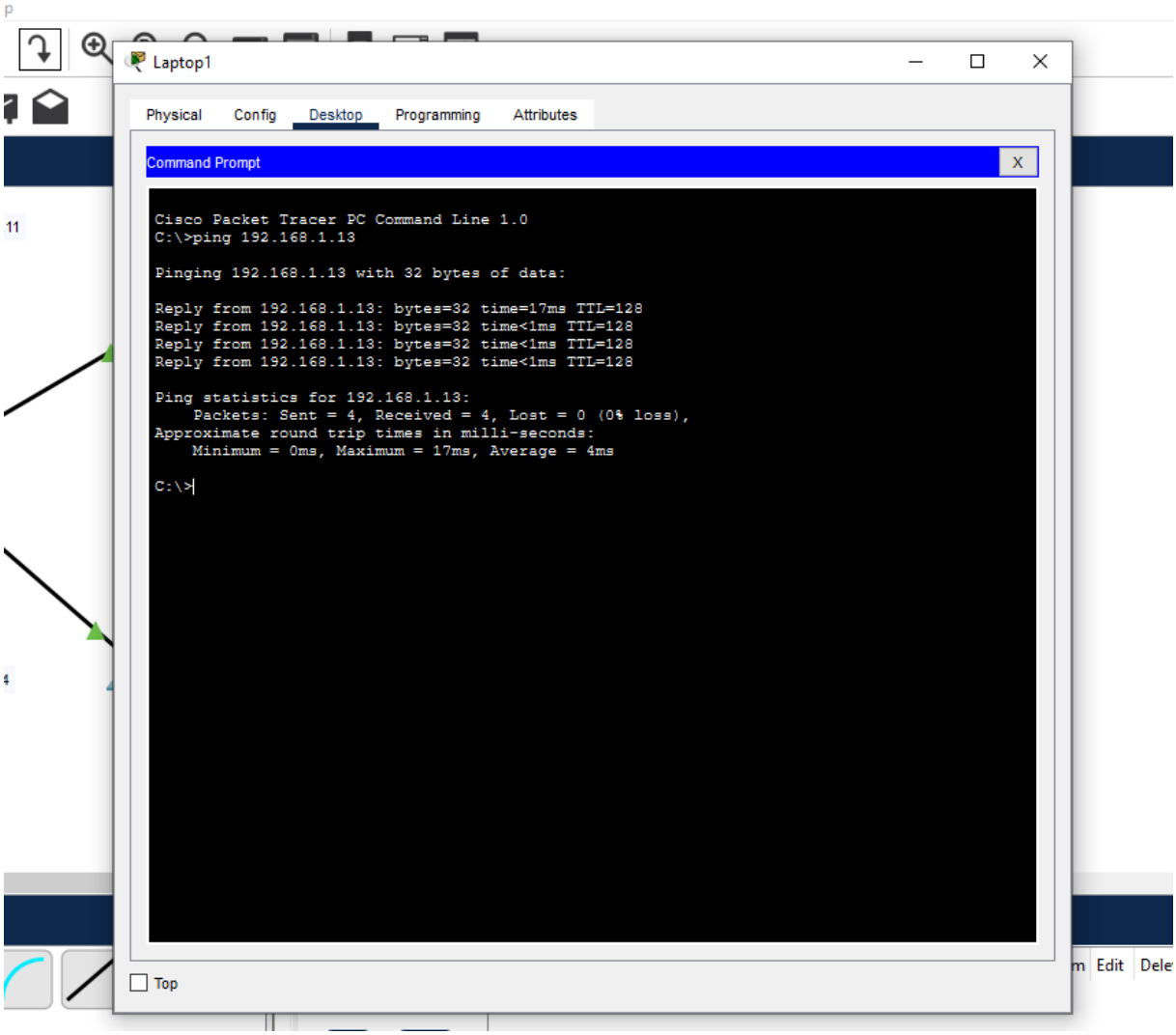
3. Reproduisez cette topologie en configurant les adresses IPv4 indiquées dans le tableau ci-dessous, puis vérifiez la connectivité

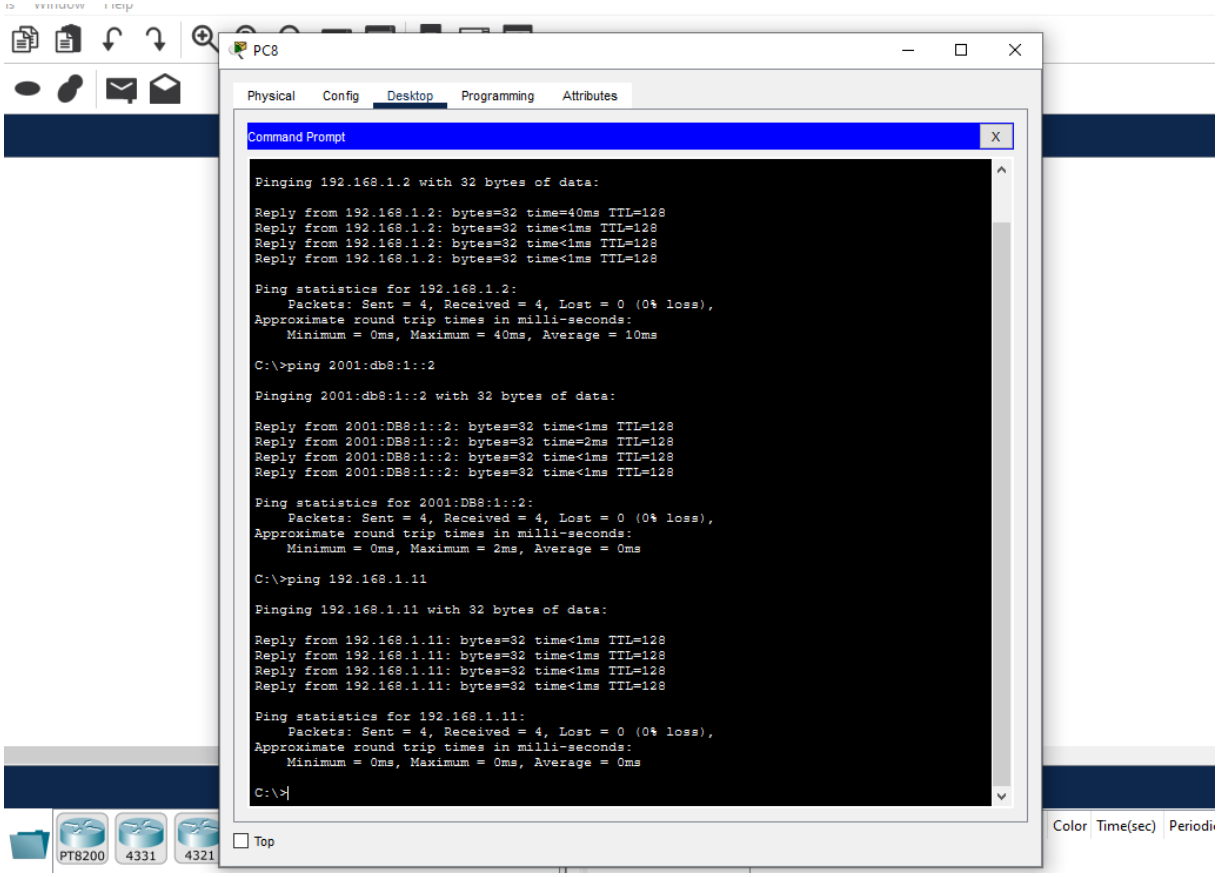



Help
PC8
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
68.1.11
3.1.14
Edit


Laptop1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::290:2BFF:FEB0:5BED
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
Edit
Delete

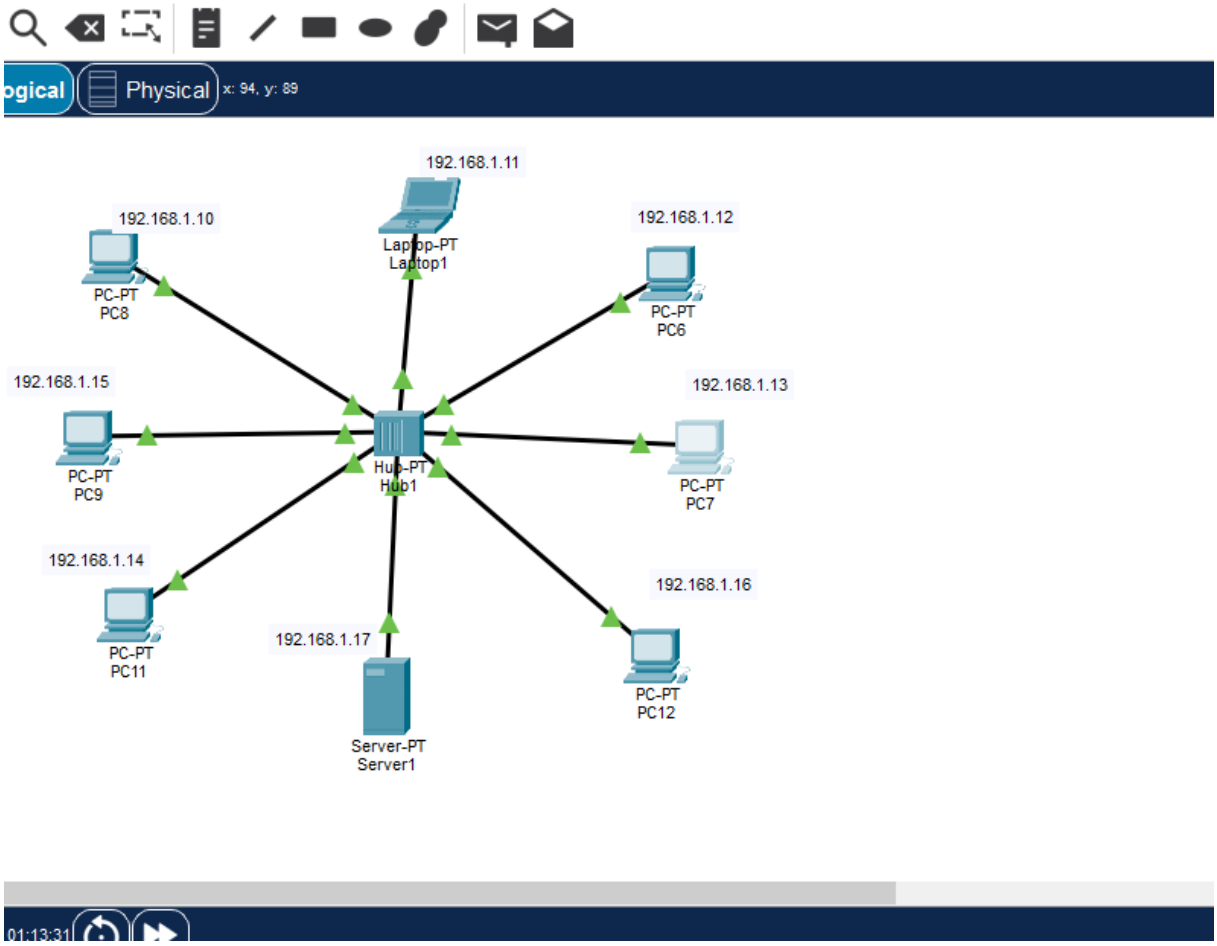
Testons la connectivité en utilisant la commande



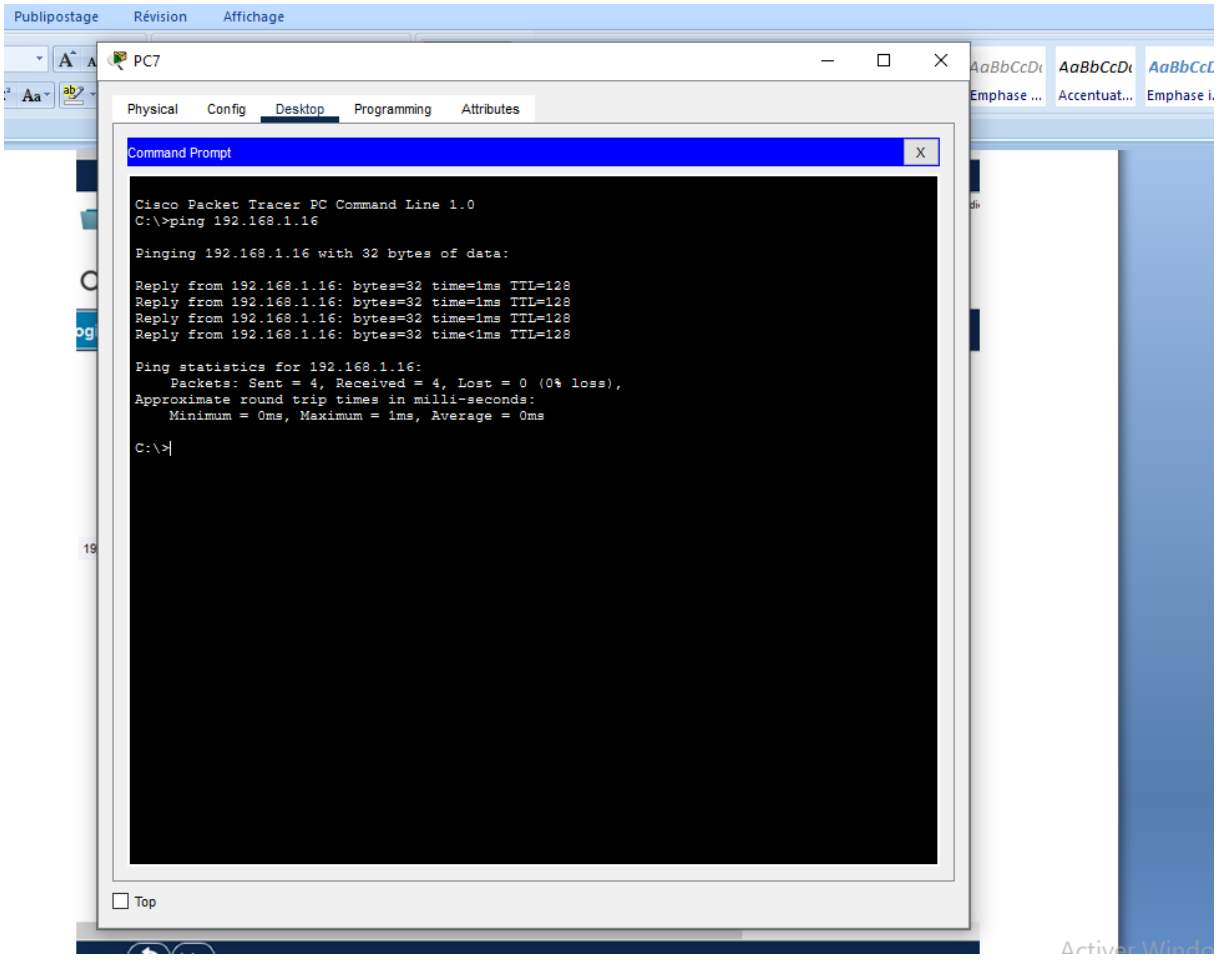


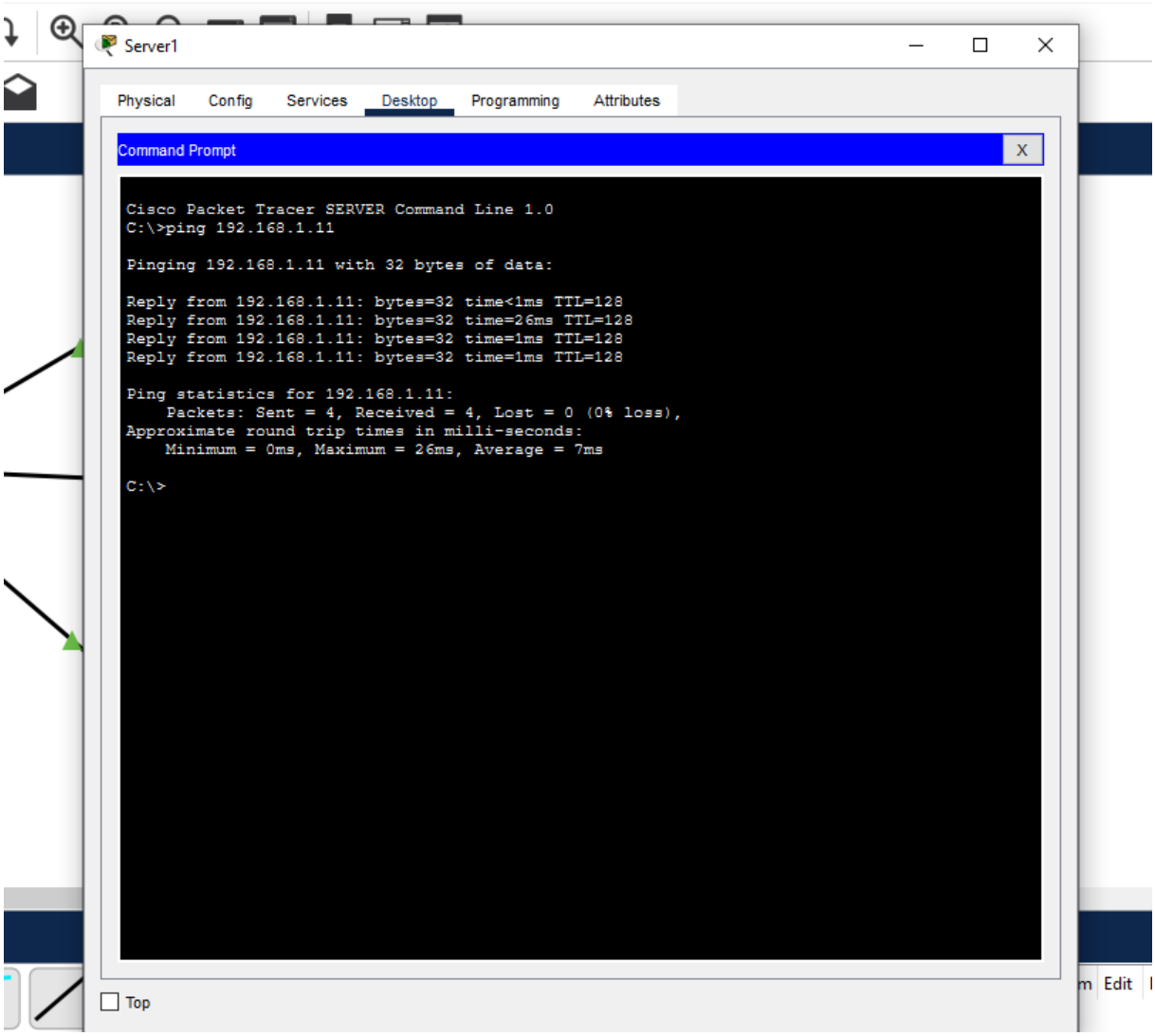
4. Reproduisez cette topologie en configurant les adresses IPv4 indiquées dans le tableau ci-dessous,

puis vérifiez la connectivité



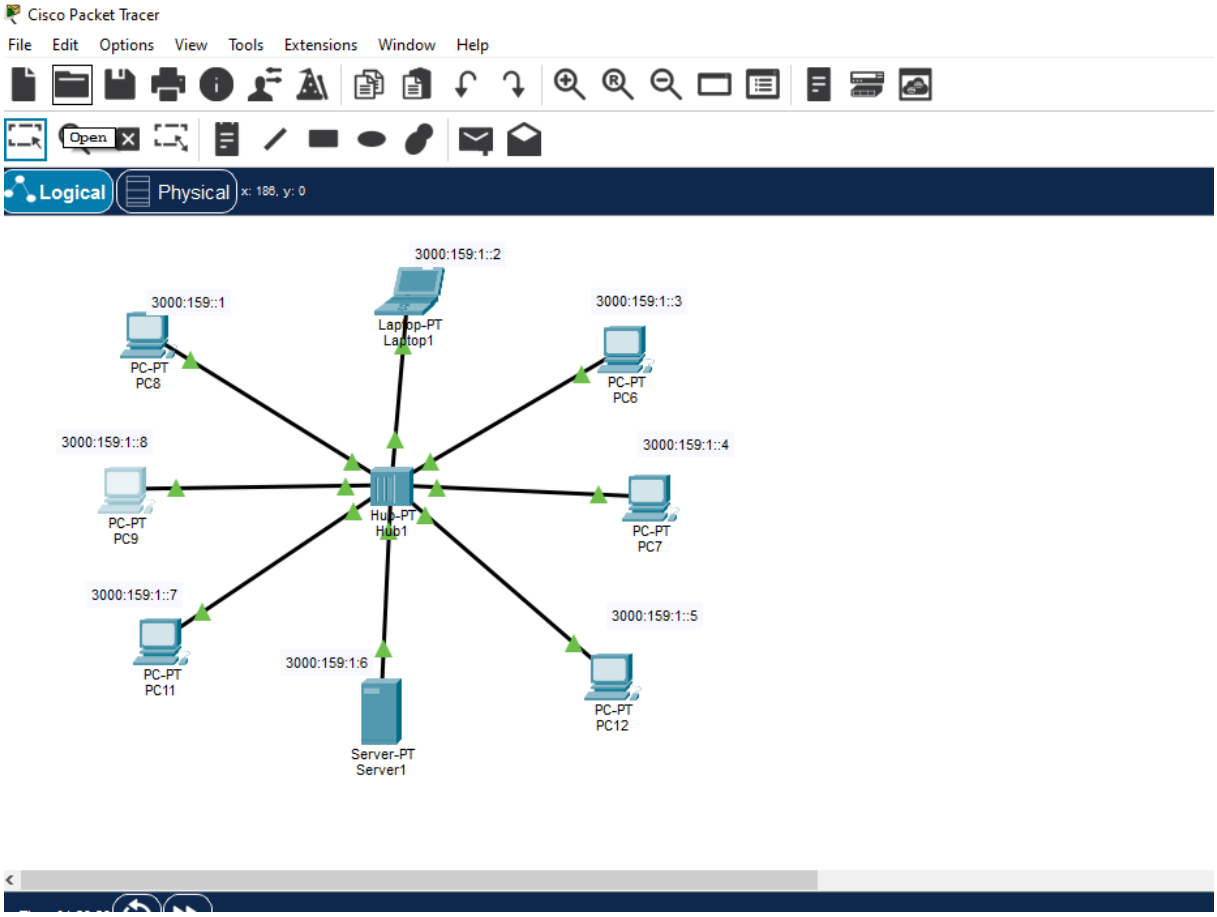
Testons la connectivité en utilisant la commande

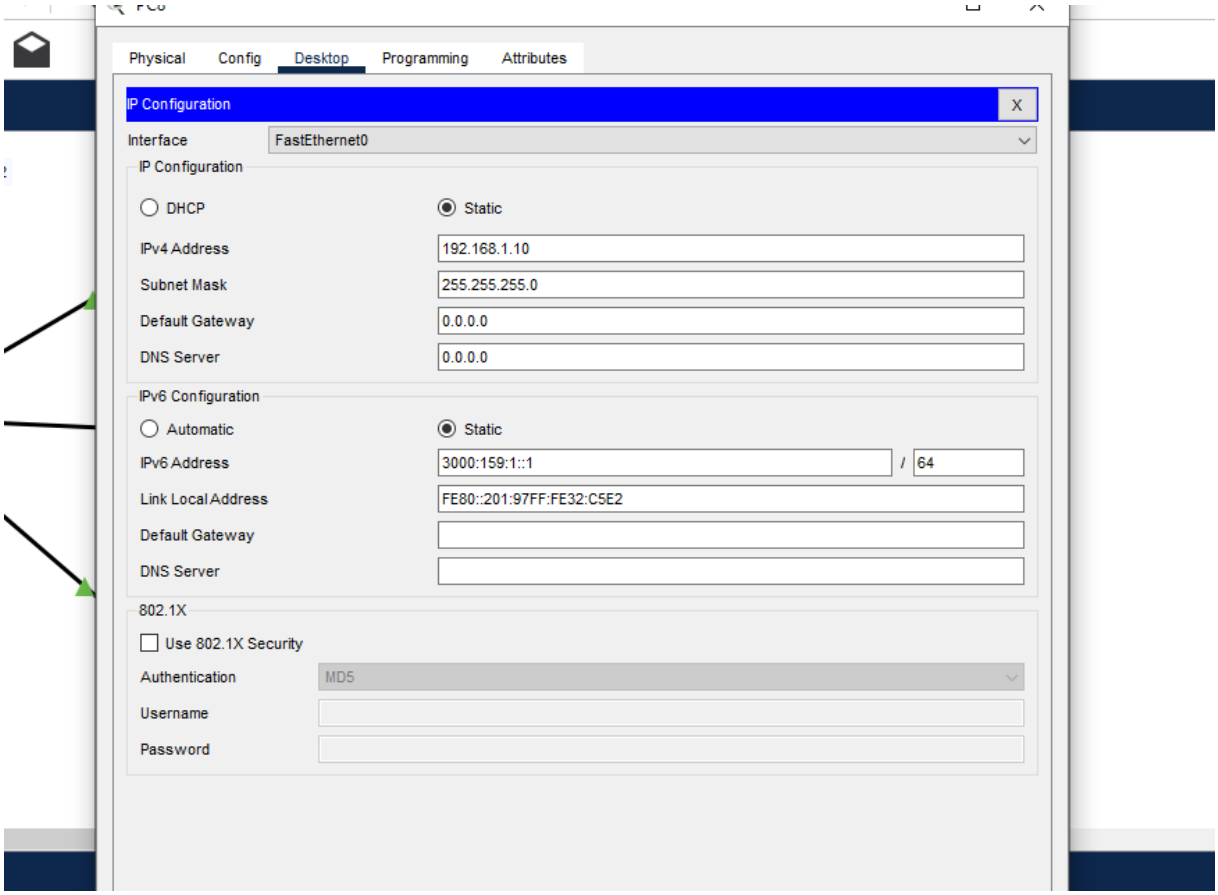


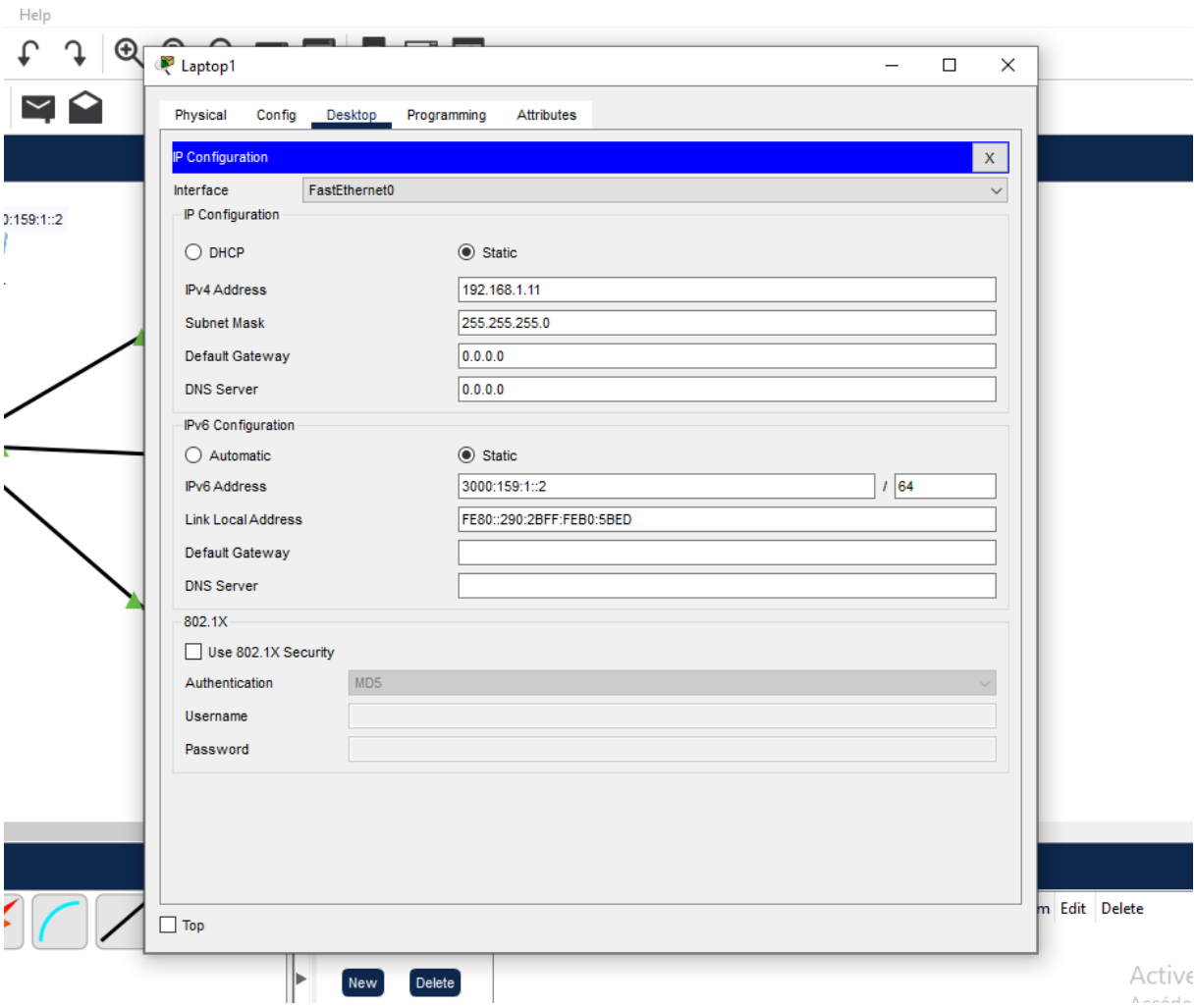


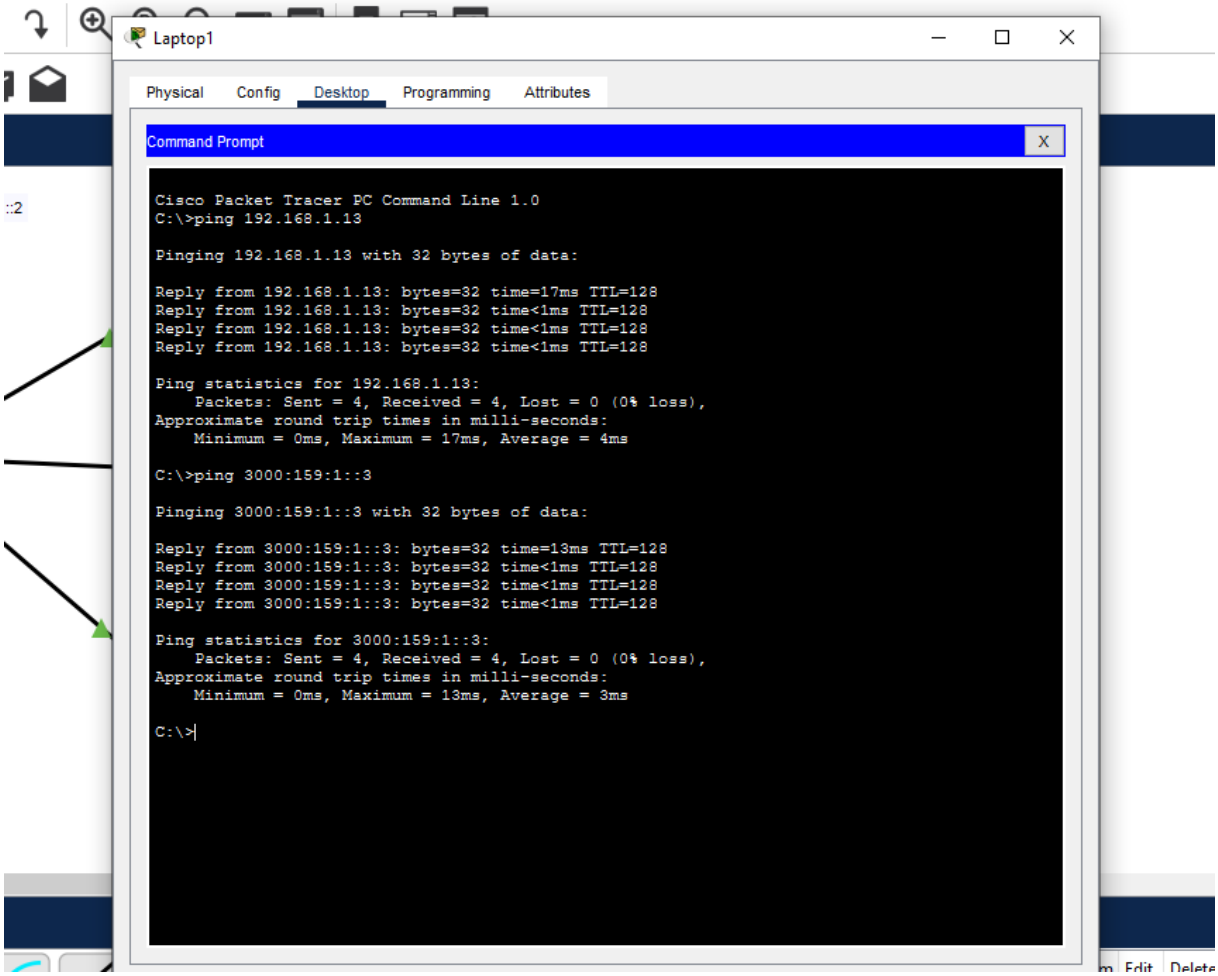
5. Reproduisez cette topologie en configurant les adresses IPv6 indiquées dans le tableau ci-dessous,

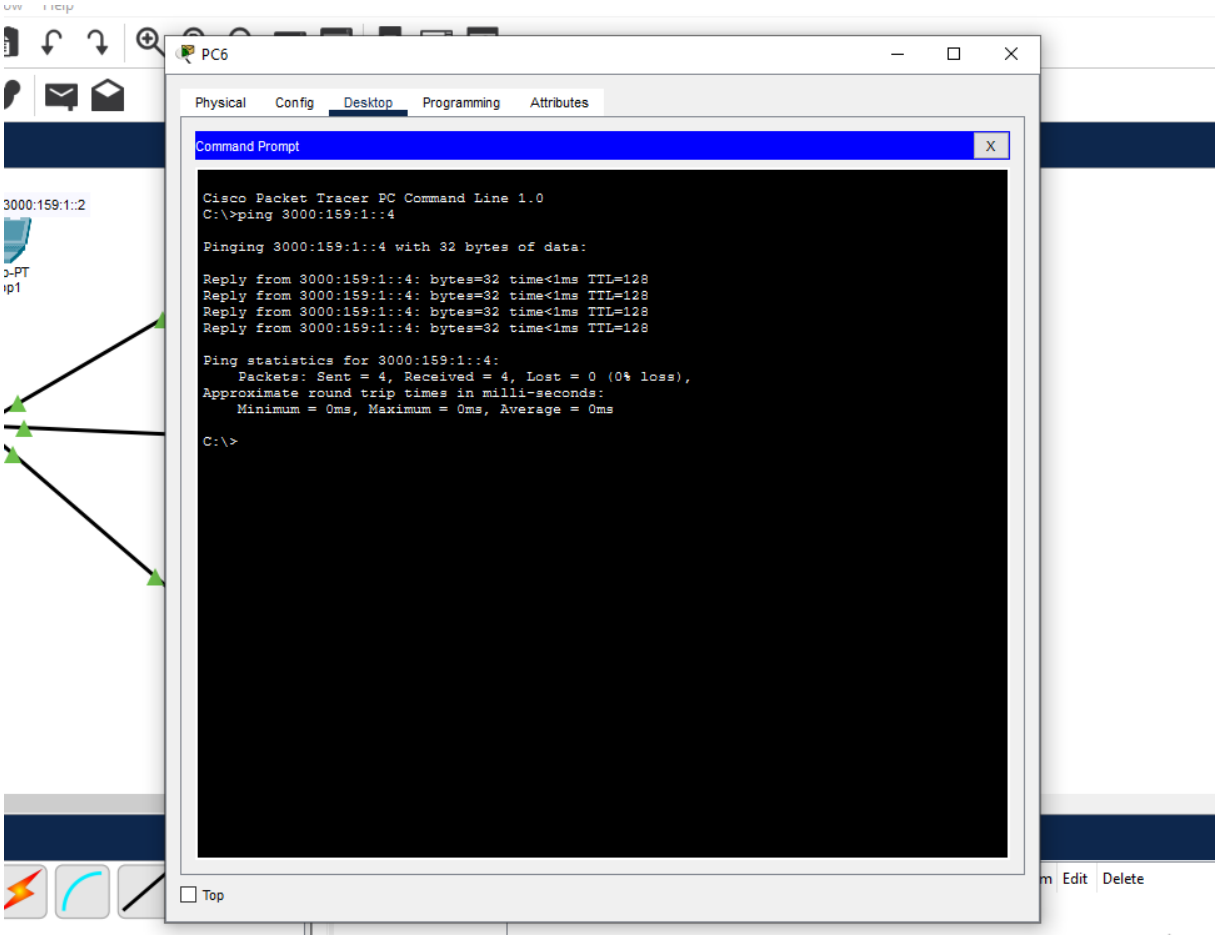
puis vérifiez la connectivité




PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.10
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::1
/
64
Link Local Address
FE80::201:97FF:FE32:C5E2
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password


Help
Laptop1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.1.11
Subnet Mask
255.255.255.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
3000:159:1::2
/
64
Link Local Address
FE80::290:2BFF:FEB0:5BED
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top
New
Delete
Edit
Delete


Laptop1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.13
Pinging 192.168.1.13 with 32 bytes of data:
Reply from 192.168.1.13: bytes=32 time=17ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Reply from 192.168.1.13: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.13:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 17ms, Average = 4ms
C:\>ping 3000:159:1::3
Pinging 3000:159:1::3 with 32 bytes of data:
Reply from 3000:159:1::3: bytes=32 time=13ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Reply from 3000:159:1::3: bytes=32 time<1ms TTL=128
Ping statistics for 3000:159:1::3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 13ms, Average = 3ms
C:\>



Conclusion

Ce TD de Réseaux I a permis de maîtriser l'attribution et la configuration des adresses IP (IPv4 et IPv6) sous Cisco Packet Tracer. Les compétences clés acquises comprennent :

Allocation d'adresses : Savoir attribuer des adresses IP valides aux différentes machines d'une topologie réseau.

Configuration pratique : Paramétrer les adresses sur les hôtes et les routeurs via l'outil de simulation.

Vérification de la connectivité : Utiliser les commandes Ping et Ping ipv6 pour valider la communication entre les équipements.